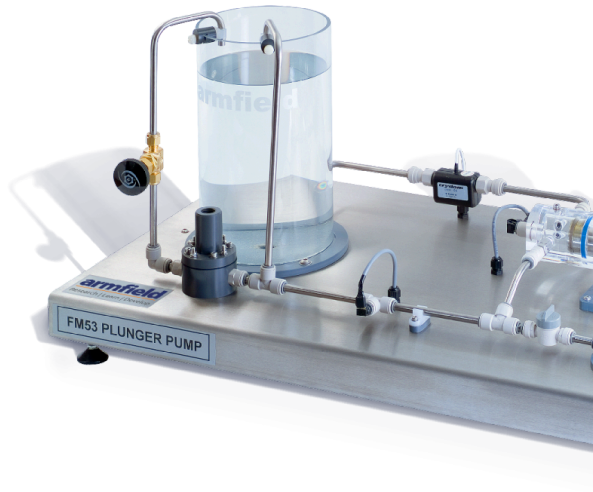




FM53 – Plunger Pump Demonstration Unit

The FM53 replicates a plunger pump as used in industry to pump small quantities of liquid at high pressures. Electronic sensors measure the instantaneous cylinder pressure within the pump, the pump displacement and the cumulative flow.

PRODUCT CODE: FM53

Categories: Civil, Educational, FM Series - Fans And Compressors, FM53 - Plunger Pump Demonstration Unit, IFT Industrial Food Technology

Tags: Armfield , Cumulative Flow , Cylinder Pressure , FM53 , High Pressure , Needle Valve , Plunger Displacement , Plunger Pump Demonstration Unit , Pulsation Dampers , Pump Displacement , Uneven Flow

DESCRIPTION

The FM53 replicates a plunger pump as used in industry to pump small quantities of liquid at high pressures. Electronic sensors measure the instantaneous cylinder pressure within the pump, the pump displacement and the cumulative flow.

The plunger or ram pump is a positive displacement pump and is used for pumping small quantities of liquid at high pressure. It is similar to a piston pump except that the sealing gland is at one end of the cylinder.

The reciprocating motion of the plunger gives an uneven flow, although the inclusion of a damping vessel can reduce this effect. Priming is unnecessary.

TECHNICAL SPECIFICATIONS

Max flow rate: 0.725 l/m typical

Max head: 4 bar

Swept volume: 15mm stroke x 32mm diameter

Pumping speed: 0 to 60 strokes/min

Motor power rating: 250W

Pressure Sensor: 0 to 100psi

Piston Sensor: -9 to 16mm

Motor Torque range: 0 to 2.1Nm

FEATURES & BENEFITS

Software

- Measurement during each pump cycle of:
 - Plunger displacement
 - Cylinder pressure
 - Pump outlet pressure
- On-line p-V diagram displays
- Measurement of volumetric efficiency
- Measuring the effect on pump performance of:
 - Sprung loading valve or needle valve
 - Adjusting the outlet loading valve
 - The inclusion of a pulsation damper vessel

Software

The ArmSOFT software enables the operator to control the fan speed 0 to 100%. Feedback from the sensors is then displayed in real time for the end user with simultaneous data logging.

The data trend is also displayed graphically in real time and can be exported to another platform such as Excel for further analysis.

Essential Accessories

Armfield IFD7

Requirements

FM53

IFD7-A: 220-240V / 1Ph / 50Hz

IFD7-G: 220-240V / 1Ph / 60Hz

Shipping Specifications

Volume: 1.0m³

Gross weight: 120kg

Dimensions

Length: 0.88m

Width: 0.51m

Height: 0.41m

Order Codes

FM53

IFD7-A: 220-240V / 1Ph / 50Hz

IFD7-G: 220-240V / 1Ph / 60Hz



Armfield Limited. Unit 10 Headlands Business Park, Salisbury Road, Ringwood, Hampshire BH24 3PB.
England. Tel: +44(0) 1425 478781